Week 4

Course Level Objectives:

- Recognize, examine, and interpret the basic principles of describing and presenting data. (CLO 2)
- Solve linear regression and correlation problems. (CLO 7)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compute the sample standard deviation of a small dataset by hand. (CLO 2) **2.** Compute a z-score for a data value. (CLO 2) **3.** Use z-scores to compare values measured on different scales or in different groups. (CLO 2) **4.** Compute the five-number summary, range, and IQR of a dataset. (CLO 2) **5.** Apply the 1.5×IQR rule to identify outliers. (CLO 2) **6.** Construct a boxplot, marking any outliers. (CLO 2)	e-Text: 2.3 One Quantitative Variable: Using Percentiles Interactive Lecture: 2.3 One Quantitative Variable: Using Percentiles Slides: 2.3 One Quantitative Variable: Using Percentiles	Participation: One Quantitative Variable (Kahoot!) – Tue: review of 2.2 Participation: Using Percentiles – Your Turn: Comparing Across Groups (heights of men and women with z-scores) Participation: Box Plots and Outliers – five-number summary, 1.5×IQR rule	Online Homework: 2.3 One Quantitative Variable: Using Percentiles (due Sun Sep 20, 11:59 PM) Take-Home Quiz 2 (due Tue Sep 22, 11:59 PM) – covers 2.1–2.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Kahoot! Zoom Respondus
7. Compare a quantitative variable across groups using side-by-side boxplots. (CLO 2) **8.** Compute and interpret a difference in means. (CLO 2) **9.** Describe the direction, form, and strength of the relationship in a scatterplot. (CLO 7) **10.** Estimate the correlation r from a scatterplot. (CLO 7) **11.** Use technology to compute the correlation r. (CLO 7) **12.** Explain how outliers, nonlinear patterns, and the absence of causation limit the interpretation of r. (CLO 7)	e-Text: 2.4 Two Variable Relationships Slides: 2.4 Two Variable Relationships	Examples: tea vs. coffee boxplots and difference in means; NBA rebounds vs. free-throw % scatterplot; find r with and without an outlier Your Turn: Tea vs. Coffee: Descriptive Statistics (difference in means; can we conclude causation?)	Online Homework: 2.4 Two Variable Relationships (due Sun Sep 20, 11:59 PM) Take-Home Quiz 3 (due Tue Oct 6, 11:59 PM) – covers 2.4–2.5 (objectives 9–12) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS Blackboard Zoom Respondus